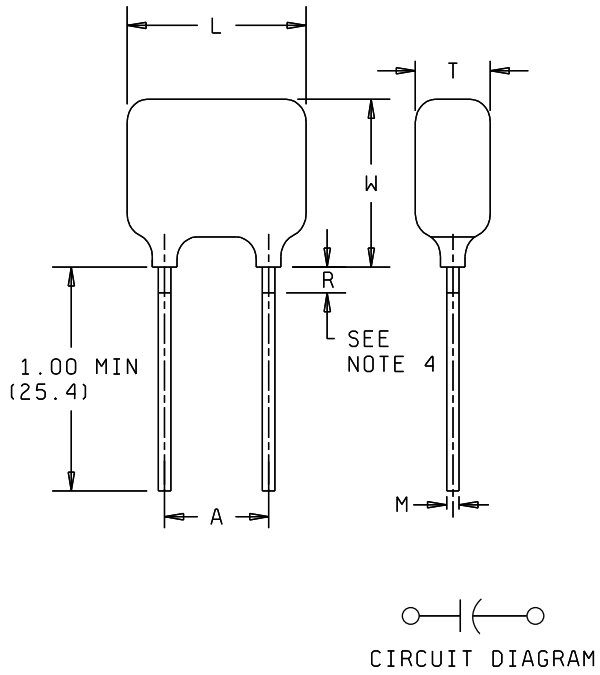
INCH-POUND
MIL-PRF-39001/5E
w/AMENDMENT 1
15 March 2019
SUPERSEDING
MIL-PRF-39001/5E
18 February 2009

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CMR03, CMR04, CMR05, CMR06, CMR07, AND CMR08

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39001.



NOTES:
1. Dimensions are in inches (see table I).
2. Millimeters are in parenthesis (see table I). Metric equivalents are given for general information only.
4. The "R" dimension may not be solderable as it may be covered by a clear epoxy or resinous coating.

FIGURE 1. Styles CMR03, CMR04, CMR05, CMR06, CMR07, and CMR08 capacitors.

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TABLE I. Dimensions.

STYLE CMR03												
Capacitance (pF)	300 V			100 V			50 V			300 V, 100 V, and 50 V		
	L	W	T	L	W	T	L	W	T	R MAX	A spacing	M DIA
	MAXIMUM			MAXIMUM			MAXIMUM				±.031 (0.79)	±.002 (0.05)
1 to 12 inclusive	.270 (6.86)	.190 (4.83)	.110 (2.79)							.078 (1.98)	.120 (3.05)	.016 (0.41)
15	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)				.078 (1.98)	.120 (3.05)	.016 (0.41)
18 and 20	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)				.078 (1.98)	.120 (3.05)	.016 (0.41)
22 and 24	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)	.078 (1.98)	.120 (3.05)	.016 (0.41)
27	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)	.078 (1.98)	.120 (3.05)	.016 (0.41)
30 and 33	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)	.078 (1.98)	.120 (3.05)	.016 (0.41)
36	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)	.078 (1.98)	.120 (3.05)	.016 (0.41)
39	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
43	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
47 and 51	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
56	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
62	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
68	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
75 and 82	.270 (6.86)	.230 (5.84)	.160 (4.06)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.078 (1.98)	.120 (3.05)	.016 (0.41)
91	.270 (6.86)	.230 (5.84)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.078 (1.98)	.120 (3.05)	.016 (0.41)
100 and 110	.270 (6.86)	.240 (6.10)	.180 (4.57)	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.078 (1.98)	.120 (3.05)	.016 (0.41)
120	.270 (6.86)	.250 (6.35)	.190 (4.83)	.270 (6.86)	.220 (5.59)	.160 (4.06)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.078 (1.98)	.120 (3.05)	.016 (0.41)
130				.270 (6.86)	.230 (5.84)	.160 (4.06)	.270 (6.86)	.210 (5.33)	.130 (3.30)	.078 (1.98)	.120 (3.05)	.016 (0.41)
150				.270 (6.86)	.230 (5.84)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.078 (1.98)	.120 (3.05)	.016 (0.41)
160				.270 (6.86)	.240 (6.10)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.078 (1.98)	.120 (3.05)	.016 (0.41)
170 and 180				.270 (6.86)	.240 (6.10)	.180 (4.57)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.078 (1.98)	.120 (3.05)	.016 (0.41)
200				.270 (6.86)	.250 (6.35)	.190 (4.83)	.270 (6.86)	.220 (5.59)	.150 (3.81)	.078 (1.98)	.120 (3.05)	.016 (0.41)
220							.270 (6.86)	.220 (5.59)	.150 (3.81)	.078 (1.98)	.120 (3.05)	.016 (0.41)
240							.270 (6.86)	.220 (5.59)	.160 (4.06)	.078 (1.98)	.120 (3.05)	.016 (0.41)
270							.270 (6.86)	.230 (5.84)	.160 (4.06)	.078 (1.98)	.120 (3.05)	.016 (0.41)
300							.270 (6.86)	.230 (5.84)	.170 (4.32)	.078 (1.98)	.120 (3.05)	.016 (0.41)
330 and 360							.270 (6.86)	.240 (6.10)	.180 (4.57)	.078 (1.98)	.120 (3.05)	.016 (0.41)
390 and 400							.270 (6.86)	.250 (6.35)	.190 (4.83)	.078 (1.98)	.120 (3.05)	.016 (0.41)

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TABLE I. Dimensions - Continued.

Capacitance (pF)	L		T		W max	R max	A spacing ± .031 (0.79)	M Dia ±.002 (0.05)
	Min	Max	Min	Max				
STYLE CMR04								
1 to 24, inclusive	.300 (7.62)	.360 (9.14)	.100 (2.54)	.190 (4.83)	.330 (8.38)	.125 (3.18)	.150 (3.81)	.016 (0.41)
27	.300 (7.62)	.370 (9.40)	.100 (2.54)	.190 (4.83)	.330 (8.38)	.125 (3.18)	.150 (3.81)	.016 (0.41)
30 to 68, inclusive	.300 (7.62)	.370 (9.40)	.100 (2.54)	.190 (4.83)	.340 (8.64)	.125 (3.18)	.150 (3.81)	.016 (0.41)
75	.300 (7.62)	.370 (9.40)	.110 (2.79)	.200 (5.08)	.340 (8.64)	.125 (3.18)	.150 (3.81)	.016 (0.41)
82	.300 (7.62)	.370 (9.40)	.110 (2.79)	.200 (5.08)	.350 (8.89)	.125 (3.18)	.150 (3.81)	.016 (0.41)
91 and 100	.300 (7.62)	.370 (9.40)	.110 (2.79)	.200 (5.08)	.350 (8.89)	.125 (3.18)	.150 (3.81)	.016 (0.41)
110 and 120	.300 (7.62)	.380 (9.65)	.110 (2.79)	.200 (5.08)	.350 (8.89)	.125 (3.18)	.150 (3.81)	.016 (0.41)
130	.300 (7.62)	.380 (9.65)	.110 (2.79)	.200 (5.08)	.360 (9.14)	.125 (3.18)	.150 (3.81)	.016 (0.41)
150 and 160	.300 (7.62)	.380 (9.65)	.120 (3.05)	.210 (5.33)	.360 (9.14)	.125 (3.18)	.150 (3.81)	.016 (0.41)
180	.300 (7.62)	.390 (9.91)	.120 (3.05)	.210 (5.33)	.370 (9.40)	.125 (3.18)	.150 (3.81)	.016 (0.41)
220 to 390, inclusive	.300 (7.62)	.390 (9.91)	.140 (3.56)	.220 (5.59)	.380 (9.65)	.125 (3.18)	.150 (3.81)	.016 (0.41)
STYLE CMR05								
1 to 62, inclusive	.400 (10.16)	.450 (11.43)	.090 (2.29)	.170 (4.32)	.360 (9.14)	.125 (3.18)	.225 (5.72)	.025 (0.64)
68 to 82, inclusive	.400 (10.16)	.450 (11.43)	.095 (2.41)	.180 (4.57)	.360 (9.14)	.125 (3.18)	.225 (5.72)	.025 (0.64)
91 and 100	.400 (10.16)	.460 (11.68)	.095 (2.41)	.180 (4.57)	.360 (9.14)	.125 (3.18)	.225 (5.72)	.025 (0.64)
110 to 130, inclusive	.400 (10.16)	.460 (11.68)	.095 (2.41)	.180 (4.57)	.370 (9.40)	.125 (3.18)	.225 (5.72)	.025 (0.64)
150 to 180, inclusive	.400 (10.16)	.460 (11.68)	.110 (2.79)	.190 (4.83)	.370 (9.40)	.125 (3.18)	.225 (5.72)	.025 (0.64)
200	.400 (10.16)	.460 (11.68)	.110 (2.79)	.190 (4.83)	.380 (9.65)	.125 (3.18)	.225 (5.72)	.025 (0.64)
220 and 240	.400 (10.16)	.460 (11.68)	.110 (2.79)	.200 (5.08)	.380 (9.65)	.125 (3.18)	.225 (5.72)	.025 (0.64)
270 to 330, inclusive	.400 (10.16)	.470 (11.94)	.110 (2.79)	.210 (5.33)	.390 (9.91)	.125 (3.18)	.225 (5.72)	.025 (0.64)
360 and 390	.400 (10.16)	.470 (11.94)	.120 (3.05)	.220 (5.59)	.400 (10.16)	.125 (3.18)	.225 (5.72)	.025 (0.64)
STYLE CMR06								
430 and 470	.550 (13.97)	.640 (16.26)	.090 (2.29)	.200 (5.08)	.510 (12.95)	.141 (3.58)	.350 (8.89)	.032 (0.81)
510 to 620, inclusive	.560 (14.22)	.650 (16.51)	.090 (2.29)	.200 (5.08)	.510 (12.95)	.141 (3.58)	.350 (8.89)	.032 (0.81)
680 to 910, inclusive	.560 (14.22)	.650 (16.51)	.100 (2.54)	.210 (5.33)	.510 (12.95)	.141 (3.58)	.350 (8.89)	.032 (0.81)
1000 and 1100	.560 (14.22)	.650 (16.51)	.110 (2.79)	.220 (5.59)	.520 (13.21)	.141 (3.58)	.350 (8.89)	.032 (0.81)
1200 and 1300	.570 (14.48)	.660 (16.76)	.110 (2.79)	.220 (5.59)	.520 (13.21)	.141 (3.58)	.350 (8.89)	.032 (0.81)
1500	.570 (14.48)	.660 (16.76)	.120 (3.05)	.230 (5.84)	.520 (13.21)	.141 (3.58)	.350 (8.89)	.032 (0.81)
1600	.570 (14.48)	.660 (16.76)	.120 (3.05)	.230 (5.84)	.530 (13.46)	.141 (3.58)	.350 (8.89)	.032 (0.81)
1800 and 2000	.580 (14.73)	.670 (17.02)	.130 (3.30)	.240 (6.10)	.530 (13.46)	.141 (3.58)	.350 (8.89)	.032 (0.81)
2200	.580 (14.73)	.670 (17.02)	.140 (3.56)	.250 (6.35)	.530 (13.46)	.141 (3.58)	.350 (8.89)	.032 (0.81)
2400	.580 (14.73)	.670 (17.02)	.150 (3.81)	.260 (6.60)	.540 (13.72)	.141 (3.58)	.350 (8.89)	.032 (0.81)
2700	.590 (14.99)	.680 (17.27)	.160 (4.06)	.270 (6.86)	.540 (13.72)	.141 (3.58)	.350 (8.89)	.032 (0.81)
3000	.590 (14.99)	.680 (17.27)	.170 (4.32)	.280 (7.11)	.550 (13.97)	.141 (3.58)	.350 (8.89)	.032 (0.81)
3300	.590 (14.99)	.680 (17.27)	.180 (4.57)	.290 (7.37)	.550 (13.97)	.141 (3.58)	.350 (8.89)	.032 (0.81)
3600	.590 (14.99)	.680 (17.27)	.190 (4.83)	.300 (7.62)	.560 (14.22)	.141 (3.58)	.350 (8.89)	.032 (0.81)
3900	.600 (15.24)	.690 (17.53)	.200 (5.08)	.310 (7.87)	.560 (14.22)	.141 (3.58)	.350 (8.89)	.032 (0.81)
4300	.600 (15.24)	.690 (17.53)	.220 (5.59)	.330 (8.38)	.570 (14.48)	.141 (3.58)	.350 (8.89)	.032 (0.81)
4700	.610 (15.49)	.700 (17.78)	.220 (5.59)	.350 (8.89)	.580 (14.73)	.141 (3.58)	.350 (8.89)	.032 (0.81)
STYLE CMR07								
5100	.680 (17.27)	.780 (19.81)	.150 (3.81)	.280 (7.11)	.860 (21.84)	.141 (3.58)	.425 (10.80)	.040 (1.02)
5600 and 6200	.680 (17.27)	.780 (19.81)	.150 (3.81)	.290 (7.37)	.870 (22.10)	.141 (3.58)	.425 (10.80)	.040 (1.02)
6800	.680 (17.27)	.780 (19.81)	.150 (3.81)	.300 (7.62)	.870 (22.10)	.141 (3.58)	.425 (10.80)	.040 (1.02)
7500	.690 (17.53)	.790 (20.07)	.150 (3.81)	.310 (7.87)	.880 (22.35)	.141 (3.58)	.425 (10.80)	.040 (1.02)
8200	.690 (17.53)	.790 (20.07)	.150 (3.81)	.320 (8.13)	.880 (22.35)	.141 (3.58)	.425 (10.80)	.040 (1.02)
9100	.690 (17.53)	.790 (20.07)	.150 (3.81)	.330 (8.38)	.880 (22.35)	.141 (3.58)	.425 (10.80)	.040 (1.02)
10,000	.700 (17.78)	.800 (20.32)	.200 (5.08)	.340 (8.64)	.890 (22.61)	.141 (3.58)	.425 (10.80)	.040 (1.02)
11,000	.700 (17.78)	.800 (20.32)	.200 (5.08)	.350 (8.89)	.890 (22.61)	.141 (3.58)	.425 (10.80)	.040 (1.02)
12,000	.700 (17.78)	.800 (20.32)	.200 (5.08)	.360 (9.14)	.890 (22.61)	.141 (3.58)	.425 (10.80)	.040 (1.02)
13,000	.710 (18.03)	.810 (20.57)	.250 (6.35)	.370 (9.40)	.890 (22.61)	.141 (3.58)	.425 (10.80)	.040 (1.02)
15,000	.710 (18.03)	.810 (20.57)	.250 (6.35)	.390 (9.91)	.900 (22.86)	.141 (3.58)	.425 (10.80)	.040 (1.02)
16,000	.720 (18.29)	.820 (20.83)	.250 (6.35)	.410 (10.41)	.900 (22.86)	.141 (3.58)	.425 (10.80)	.040 (1.02)
18,000	.720 (18.29)	.820 (20.83)	.250 (6.35)	.430 (10.92)	.910 (23.11)	.141 (3.58)	.425 (10.80)	.040 (1.02)
20,000	.730 (18.54)	.830 (21.08)	.250 (6.35)	.450 (11.43)	.920 (23.37)	.141 (3.58)	.425 (10.80)	.040 (1.02)

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TABLE I. Dimensions - Continued.

Capacitance (pF)	L		T		W	R	A spacing ± .031 (0.79)	M Dia ±.002 (0.05)
	Min	Max	Min	Max	max	max		
STYLE CMR08								
22,000	1.300 (33.02)	1.420 (36.07)	.180 (4.57)	.310 (7.87)	.880 (22.35)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
24,000	1.310 (33.27)	1.430 (36.32)	.180 (4.57)	.320 (8.13)	.880 (22.35)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
27,000	1.320 (33.53)	1.430 (36.32)	.180 (4.57)	.330 (8.38)	.880 (22.35)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
30,000	1.330 (33.78)	1.440 (36.58)	.180 (4.57)	.350 (8.89)	.890 (22.61)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
33,000	1.330 (33.78)	1.440 (36.58)	.180 (4.57)	.360 (9.14)	.890 (22.61)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
36,000	1.340 (34.04)	1.450 (36.83)	.180 (4.57)	.380 (9.65)	.900 (22.86)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
39,000	1.340 (34.04)	1.450 (36.83)	.220 (5.59)	.400 (10.16)	.900 (22.86)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
43,000	1.350 (34.29)	1.460 (37.08)	.225 (5.72)	.420 (10.67)	.910 (23.11)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
47,000	1.360 (34.54)	1.470 (37.34)	.250 (6.35)	.450 (11.43)	.910 (23.11)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
51,000	1.370 (34.80)	1.480 (37.59)	.250 (6.35)	.470 (11.94)	.920 (23.37)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
56,000	1.350 (34.29)	1.460 (37.08)	.225 (5.72)	.420 (10.67)	.910 (23.11)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
62,000	1.360 (34.54)	1.470 (37.34)	.250 (6.35)	.450 (11.43)	.920 (23.37)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
68,000	1.370 (34.80)	1.480 (37.59)	.260 (6.60)	.470 (11.94)	.920 (23.37)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
75,000	1.370 (34.54)	1.470 (37.34)	.240 (6.10)	.440 (11.18)	.910 (23.11)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
82,000	1.370 (34.80)	1.480 (37.59)	.310 (7.87)	.460 (11.68)	.920 (23.37)	.141 (3.58)	1.050 (26.67)	.040 (1.02)
91,000	1.390 (35.31)	1.500 (38.10)	.340 (8.64)	.500 (12.70)	.940 (23.88)	.141 (3.58)	1.050 (26.67)	.040 (1.02)

REQUIREMENTS:

Dimension: See figure 1 and table I.

Case: Resin (dipped).

Capacitance value: See table II.

Capacitance tolerance: D (± 0.5 F); F (±1%); G (±2%); and J (±5%) as shown in table II.

DC rated voltage: Y (50 V), A (100 V), C (300 V), and D (500 V) as shown in table II.

Operating temperature range: O (-55° to 125°C) and P (-55° to +150°C) as shown in table II.

Characteristic: C, E, and F as specified in MIL-PRF-39001.

Product level: C, M, P, R, and S as specified in MIL-PRF-39001.

Barometric pressure: In accordance with MIL-STD-202-105, condition D (100,000 ft). Test potential: 100 percent of rated dc voltage for style CMR04, 500 volts; 150 percent for lesser voltages and all other styles.

Insulation resistance (IR): In accordance with MIL-STD-202-302, test condition A: (Minimum requirements):

At +25°C:
0 to 10,000 pF..... 100,000 megohms
10,000 pF and greater ..... 1,000 megohm-microfarads

At +125°C:
0 to 3,300 pF..... 10,000 megohms
3,300 pF and greater ..... 33 megohm-microfarads

At +150°C
0 to 1,500 pF..... 5,000 megohms
1,500 pF and greater ..... 7.5 megohm-microfarads

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Dissipation factor (DF): See MIL-PRF-39001.

High voltage stabilization: 200 percent of rated dc voltage for 48 hours at a minimum temperature of +125°C or +150°C, as applicable.

Thermal shock: In accordance with MIL-PRF-39001, except high temperature shall be as follows:
Symbol O - +125°C.
Symbol P - +150°C.

Immersion: In accordance with MIL-PRF-39001, except ΔCap. shall not exceed ±1 percent or 1pF,whichever is greater, of reference measurement.

Terminal strength:

Pull test: For style CMR03 and CMR04, pull shall be applied in the direction of the leads.

Moisture resistance: In accordance with MIL-PRF-39001, except ΔCap. shall not exceed ±1 percent or 1 pF, whichever is greater, of reference measurement.

Life:

Forty eight hours, minimum, at -55°C +0°C, -2°C immediately followed by qualification or quality conformance test, as applicable.

Qualification test: 2,000 hours at applicable high test temperature with 100 percent or 150 percent of rated dc voltage applied, as applicable.

DWV: Meet initial requirement.

IR: Meet initial requirement.

ΔCap.: Shall not exceed ±1 percent or 1 pF, whichever is greater, or the initial measurement.

DF: Not greater than 150 percent of initial requirement.

Quality conformance test: 2,000 hours (M, P, R, and S FRLs) at applicable high test temperature and 150 percent of rated dc voltage; requirements as specified for qualification.

Resistance to soldering heat: In accordance with MIL-PRF-39001, except ΔCap. shall not exceed ±1 percent or 1 pF, whichever is greater, of the initial measurement.

Marking: Capacitors shall be marked as indicated in examples on figure 2.

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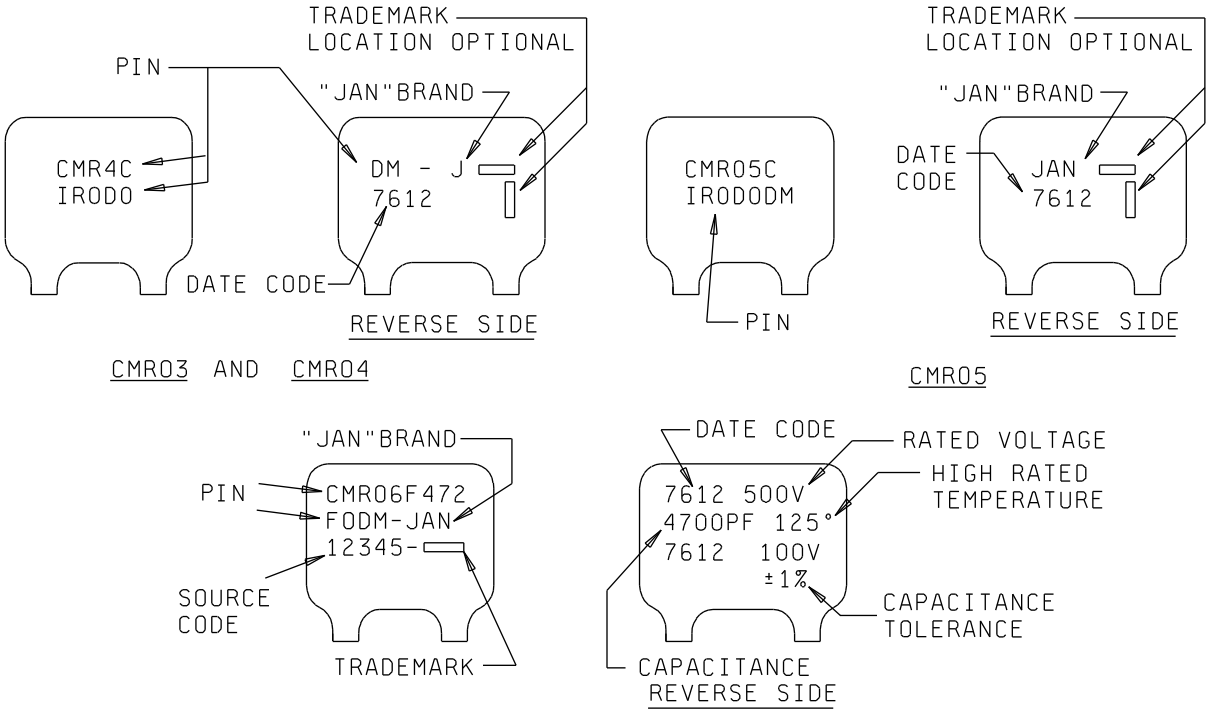


FIGURE 2. Marking examples.

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TABLE II. Capacitor characteristics.

STYLE CMR03				
Part or identifying number (PIN) 1/	Capacitance value (pF)	DC rated voltage (volts)	Operating temperature range	Capacitance tolerance available
CMR03C1R0DOC -	1.0	300	O	D
CMR03C1R5DOC -	1.5	300	O	D
CMR03C2R0DOC -	2.0	300	O	D
CMR03C2R5DOC -	2.5	300	O	D
CMR03C3R0DOC -	3.0	300	O	D
CMR03C3R5DOC -	3.5	300	O	D
CMR03C4R0DOC -	4.0	300	O	D
CMR03C4R5DOC -	4.5	300	O	D
CMR03C5R0DOC -	5.0	300	O	D
CMR03C6R0DOC -	6.0	300	O	D
CMR03C7R0DOC -	7.0	300	O	D
CMR03C8R0DOC -	8.0	300	O	D
CMR03C9R0DOC -	9.0	300	O	D
CMR03C100DOC -	10	300	O	D
CMR03C110DOC -	11	300	O	D
CMR03C120DOC -	12	300	O	D
CMR03C150DO - -	15	300, 100	O	D
CMR03C180DO - -	18	300, 100	O	D
CMR03E200DO - -	20	300, 100	O	D
CMR03E220DO - -	22	300, 100, 50	O	D
CMR03E240DO - -	24	300, 100, 50	O	D
CMR03E270 - O - -	27	300, 100, 50	O	G, J
CMR03E300 - O - -	30	300, 100, 50	O	G, J
CMR03E330 - O - -	33	300, 100, 50	O	G, J
CMR03E360 - O - -	36	300, 100, 50	O	G, J
CMR03E390 - O - -	39	300, 100, 50	O	G, J
CMR03E430 - O - -	43	300, 100, 50	O	G, J
CMR03E470 - O - -	47	300, 100, 50	O	G, J
CMR03E510 - O - -	51	300, 100, 50	O	G, J
CMR03E560 - O - -	56	300, 100, 50	O	G, J
CMR03E620 - O - -	62	300, 100, 50	O	G, J
CMR03E680 - O - -	68	300, 100, 50	O	G, J
CMR03E750 - O - -	75	300, 100, 50	O	G, J
CMR03E820 - O - -	82	300, 100, 50	O	G, J
CMR03F910 - O - -	91	300, 100, 50	O	G, J
CMR03F101 - O - -	100	300, 100, 50	O	G, J
CMR03F111 - O - -	110	300, 100, 50	O	G, J
CMR03F121 - O - -	120	300, 100, 50	O	G, J
CMR03F131 - O - -	130	100, 50	O	G, J
CMR03F151 - O - -	150	100, 50	O	G, J
CMR03F161 - O - -	160	100, 50	O	G, J
CMR03F171 - O - -	170	100, 50	O	G, J
CMR03F181 - O - -	180	100, 50	O	G, J
CMR03F201 - O - -	200	100, 50	O	G, J
CMR03F221 - OY -	220	50	O	G, J
CMR03F241 - OY -	240	50	O	G, J
CMR03F271 - OY -	270	50	O	G, J
CMR03F301 - OY -	300	50	O	G, J
CMR03F331 - OY -	330	50	O	G, J
CMR03F361 - OY -	360	50	O	G, J
CMR03F391 - OY -	390	50	O	G, J
CMR03F401 - OY -	400	50	O	G, J

See footnote at end of table.

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TABLE II. Capacitor characteristics - Continued.

STYLE CMR04				
PIN 1/	Capacitance value (pF)	DC rated voltage (volts)	Operating temperature range	Capacitance tolerance available
CMR04C1R0D - D -	1.0	500	O, P	D
CMR04C1R5D - D -	1.5	500	O, P	D
CMR04C2R0D - D -	2.0	500	O, P	D
CMR04C2R5D - D -	2.5	500	O, P	D
CMR04C3R0D - D -	3.0	500	O, P	D
CMR04C3R5D - D -	3.5	500	O, P	D
CMR04C4R0D - D -	4.0	500	O, P	D
CMR04C4R5D - D -	4.5	500	O, P	D
CMR04C5R0D - D -	5.0	500	O, P	D
CMR04C6R0D - D -	6.0	500	O, P	D
CMR04C7R0D - D -	7.0	500	O, P	D
CMR04C8R0D - D -	8.0	500	O, P	D
CMR04C9R0D - D -	9.0	500	O, P	D
CMR04C100D - D -	10	500	O, P	D
CMR04C110D - D -	11	500	O, P	D
CMR04C120J - D -	12	500	O, P	J
CMR04C150J - D -	15	500	O, P	J
CMR04C180J - D -	18	500	O, P	J
CMR04E200J - D -	20	500	O, P	J
CMR04E220J - D -	22	500	O, P	J
CMR04E240J - D -	24	500	O, P	J
CMR04E270 - - D -	27	500	O, P	G, J
CMR04E300 - - D -	30	500	O, P	G, J
CMR04E330 - - D -	33	500	O, P	G, J
CMR04E360 - - D -	36	500	O, P	G, J
CMR04E390 - - D -	39	500	O, P	G, J
CMR04E430 - - D -	43	500	O, P	G, J
CMR04E470 - - D -	47	500	O, P	G, J
CMR04E510 - - D -	51	500	O, P	F, G, J
CMR04E560 - - D -	56	500	O, P	F, G, J
CMR04E620 - - D -	62	500	O, P	F, G, J
CMR04E680 - - D -	68	500	O, P	F, G, J
CMR04E750 - - D -	75	500	O, P	F, G, J
CMR04E820 - - D -	82	500	O, P	F, G, J
CMR04F910 - - D -	91	500	O, P	F, G, J
CMR04F101 - - D -	100	500	O, P	F, G, J
CMR04F111 - - D -	110	500	O, P	F, G, J
CMR04F121 - - D -	120	500	O, P	F, G, J
CMR04F131 - - D -	130	500	O, P	F, G, J
CMR04F151 - - D -	150	500	O, P	F, G, J
CMR04F161 - - D -	160	500	O, P	F, G, J
CMR04F171 - - D -	170	500	O, P	F, G, J
CMR04F181 - - D -	180	500	O, P	F, G, J
CMR04F201 - - D -	200	500	O, P	F, G, J
CMR04F221 - - D -	220	500	O, P	F, G, J
CMR04F241 - - D -	240	500	O, P	F, G, J
CMR04F271 - - C -	270	300	O, P	F, G, J
CMR04F301 - - C -	300	300	O, P	F, G, J
CMR04F331 - - A -	330	100	O, P	F, G, J
CMR04F361 - - A -	360	100	O, P	F, G, J
CMR04F391 - - A -	390	100	O, P	F, G, J

See footnote at end of table.

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TABLE II. Capacitor characteristics - Continued.

STYLE CMR05				
PIN 1/	Capacitance value (pF)	DC rated voltage (volts)	Operating temperature range	Capacitance tolerance available
CMR05C1R0D - D -	1.0	500	O, P	D
CMR05C1R5D - D -	1.5	500	O, P	D
CMR05C2R0D - D -	2.0	500	O, P	D
CMR05C2R5D - D -	2.5	500	O, P	D
CMR05C3R0D - D -	3.0	500	O, P	D
CMR05C3R5D - D -	3.5	500	O, P	D
CMR05C4R0D - D -	4.0	500	O, P	D
CMR05C4R5D - D -	4.5	500	O, P	D
CMR05C5R0D - D -	5.0	500	O, P	D
CMR05C6R0D - D -	6.0	500	O, P	D
CMR05C7R0D - D -	7.0	500	O, P	D
CMR05C8R0D - D -	8.0	500	O, P	D
CMR05C9R0D - D -	9.0	500	O, P	D
CMR05C100D - D -	10	500	O, P	D
CMR05C110D - D -	11	500	O, P	D
CMR05C120J - D -	12	500	O, P	J
CMR05C150J - D -	15	500	O, P	J
CMR05C180J - D -	18	500	O, P	J
CMR05E200J - D -	20	500	O, P	J
CMR05E220J - D -	22	500	O, P	J
CMR05E240J - D -	24	500	O, P	J
CMR05E270 - - D -	27	500	O, P	G, J
CMR05E300 - - D -	30	500	O, P	G, J
CMR05E330 - - D -	33	500	O, P	G, J
CMR05E360 - - D -	36	500	O, P	G, J
CMR05E390 - - D -	39	500	O, P	G, J
CMR05E430 - - D -	43	500	O, P	G, J
CMR05E470 - - D -	47	500	O, P	G, J
CMR05E510 - - D -	51	500	O, P	F, G, J
CMR05E560 - - D -	56	500	O, P	F, G, J
CMR05E620 - - D -	62	500	O, P	F, G, J
CMR05E680 - - D -	68	500	O, P	F, G, J
CMR05E750 - - D -	75	500	O, P	F, G, J
CMR05E820 - - D -	82	500	O, P	F, G, J
CMR05F910 - - D -	91	500	O, P	F, G, J
CMR05F101 - - D -	100	500	O, P	F, G, J
CMR05F111 - - D -	110	500	O, P	F, G, J
CMR05F121 - - D -	120	500	O, P	F, G, J
CMR05F131 - - D -	130	500	O, P	F, G, J
CMR05F151 - - D -	150	500	O, P	F, G, J
CMR05F161 - - D -	160	500	O, P	F, G, J
CMR05F171 - - D -	170	500	O, P	F, G, J
CMR05F181 - - D -	180	500	O, P	F, G, J
CMR05F201 - - D -	200	500	O, P	F, G, J
CMR05F221 - - D -	220	500	O, P	F, G, J
CMR05F241 - - D -	240	500	O, P	F, G, J
CMR05F271 - - D -	270	500	O, P	F, G, J
CMR05F301 - - D -	300	500	O, P	F, G, J
CMR05F331 - - D -	330	500	O, P	F, G, J
CMR05F361 - - D -	360	500	O, P	F, G, J
CMR05F391 - - D -	390	500	O, P	F, G, J

See footnote at end of table.

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TABLE II. Capacitor characteristics - Continued.

STYLE CMR06				
PIN 1/	Capacitance value (pF)	DC rated voltage (volts)	Operating temperature range	Capacitance tolerance available
CMR06F431 - - D -	430	500	O, P	F, G, J
CMR06F471 - - D -	470	500	O, P	F, G, J
CMR06F511 - - D -	510	500	O, P	F, G, J
CMR06F561 - - D -	560	500	O, P	F, G, J
CMR06F621 - - D -	620	500	O, P	F, G, J
CMR06F681 - - D -	680	500	O, P	F, G, J
CMR06F751 - - D -	750	500	O, P	F, G, J
CMR06F821 - - D -	820	500	O, P	F, G, J
CMR06F911 - - D -	910	500	O, P	F, G, J
CMR06F102 - - D -	1,000	500	O, P	F, G, J
CMR06F112 - - D -	1,100	500	O, P	F, G, J
CMR06F122 - - D -	1,200	500	O, P	F, G, J
CMR06F132 - - D -	1,300	500	O, P	F, G, J
CMR06F152 - - D -	1,500	500	O, P	F, G, J
CMR06F162 - - D -	1,600	500	O, P	F, G, J
CMR06F182 - - D -	1,800	500	O, P	F, G, J
CMR06F202 - - D -	2,000	500	O, P	F, G, J
CMR06F222 - - D -	2,200	500	O, P	F, G, J
CMR06F242 - - D -	2,400	500	O, P	F, G, J
CMR06F272 - - D -	2,700	500	O, P	F, G, J
CMR06F302 - - D -	3,000	500	O, P	F, G, J
CMR06F332 - - D -	3,300	500	O, P	F, G, J
CMR06F362 - - D -	3,600	500	O, P	F, G, J
CMR06F392 - - D -	3,900	500	O, P	F, G, J
CMR06F432 - - D -	4,300	500	O, P	F, G, J
CMR06F472 - - D -	4,700	500	O, P	F, G, J
STYLE CMR07				
CMR07F512 - - D -	5,100	500	O, P	F, G, J
CMR07F562 - - D -	5,600	500	O, P	F, G, J
CMR07F622 - - D -	6,200	500	O, P	F, G, J
CMR07F682 - - D -	6,800	500	O, P	F, G, J
CMR07F752 - - D -	7,500	500	O, P	F, G, J
CMR07F822 - - D -	8,200	500	O, P	F, G, J
CMR07F912 - - D -	9,100	500	O, P	F, G, J
CMR07F103 - - D -	10,000	500	O, P	F, G, J
CMR07F113 - - D -	11,000	500	O, P	F, G, J
CMR07F123 - - D -	12,000	500	O, P	F, G, J
CMR07F133 - - D -	13,000	500	O, P	F, G, J
CMR07F153 - - D -	15,000	500	O, P	F, G, J
CMR07F163 - - D -	16,000	500	O, P	F, G, J
CMR07F183 - - D -	18,000	500	O, P	F, G, J
CMR07F203 - - D -	20,000	500	O, P	F, G, J

See footnote at end of table.

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TABLE II. Capacitor characteristics - Continued.

STYLE CMR08				
PIN 1/	Capacitance value (pF)	DC rated voltage (volts)	Operating temperature range	Capacitance tolerance available
CMR08F223 - - D -	22000	500	O, P	F, G, J
CMR08F243 - - D -	24000	500	O, P	F, G, J
CMR08F273 - - D -	27000	500	O, P	F, G, J
CMR08F303 - - D -	30000	500	O, P	F, G, J
CMR08F333 - - D -	33000	500	O, P	F, G, J
CMR08F363 - - D -	36000	500	O, P	F, G, J
CMR08F393 - - D -	39000	500	O, P	F, G, J
CMR08F433 - - D -	43000	500	O, P	F, G, J
CMR08F473 - - D -	47000	500	O, P	F, G, J
CMR08F513 - - D -	51000	500	O, P	F, G, J
CMR08F563 - - C -	56000	300	O, P	F, G, J
CMR08F623 - - C -	62000	300	O, P	F, G, J
CMR08F683 - - C -	68000	300	O, P	F, G, J
CMR08F753 - - A -	75000	100	O, P	F, G, J
CMR08F823 - - A -	82000	100	O, P	F, G, J
CMR08F913 - - A -	91000	100	O, P	F, G, J

1/ Complete PIN shall include additional symbols to indicate capacitance tolerance, operating temperature range, rated voltage, and product level, where applicable:

TABLE III. Substitutability data.

Substitute item:	Substitute for:
MIL-PRF-39001/5E	MIL-C-39001/5
CMR04C1R0D-D-	CMR04C010D-D-
CMR04C5R0D-D-	CMR04C050D-D-
CMR04C6R0D-D-	CMR04C060D-D-
CMR04C7R0D-D-	CMR04C070D-D-
CMR04C8R0D-D-	CMR04C080D-D-
CMR05C1R0D-D-	CMR05C010D-D-
CMR05C5R0D-D-	CMR05C050D-D-
CMR05C6R0D-D-	CMR05C060D-D-
CMR05C7R0D-D-	CMR05C070D-D-
CMR05C8R0D-D-	CMR05C080D-D-

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Referenced documents: In addition to MIL-PRF-39001, this document also references:

MIL-STD-202-105
MIL-STD-202-302

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-2019-008)

Review activities:
Army - AV, MI
Navy - MC
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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